

Workflow Summary

Repository documentation for the Higgs-portal VBF invisible-Higgs analysis

This document summarizes the computational workflow used to move from published invisible-Higgs limits to Higgs-portal vector dark matter constraints. It is intended to accompany the repository and to make the analysis chain easier to follow, verify, and update.

Core interpretation chain:

$\text{BR}(H \rightarrow \text{inv}) \rightarrow \Gamma_{\text{inv}}^{\text{max}} \rightarrow g_{hVV}^{\text{max}}(m_V) \rightarrow \sigma_{\text{SI}}^{\text{max}}(m_V)$

Main workflow stages:

1. Collect collider input limits from selected ATLAS and CMS invisible-Higgs analyses.
2. Convert observed and expected $\text{BR}(H \rightarrow \text{inv})$ limits into maximum invisible Higgs-width values.
3. Evaluate the Higgs-portal vector dark matter coupling bound over the kinematically allowed mass range $m_V < m_h/2$.
4. Generate the full coupling-bound plot and the separate threshold-region zoom.
5. Propagate the ATLAS 2022 VBF expected ± 1 sigma and ± 2 sigma ranges into coupling-space uncertainty bands.
6. Translate the collider-derived coupling bound into a spin-independent scattering limit and overlay it with LZ, XENONnT, and PandaX-4T.
7. Export final figures in PNG and PDF form for thesis use.

Repository files connected to the workflow:

Workflow stage	Main input files	Main output files
Collider inputs	datasets/collider_limits.csv; datasets/atlas2022_vbf_uncertainty.csv	tables and BR inputs for reinterpretation
BR-to-width conversion	datasets/collider_limits.csv; datasets/higgs_constants.csv	derived invisible-width values
Width-to-coupling mapping	scripts/02_width_to_coupling.py	figure5_1_evolution_overlay and figure5_2_threshold_zoom
Uncertainty bands	datasets/atlas2022_vbf_uncertainty.csv; scripts/08_uncertainty_band_propagation.py	propagated ATLAS VBF sensitivity bands
Direct-detection overlay	datasets/lz_limit.csv; datasets/xenonnnt_limit.csv; datasets/pandax4t_digitized.csv	figure5_3_sigmaSI_overlay

Scope note: This workflow is a phenomenological reinterpretation of published limits. It does not reconstruct ATLAS or CMS likelihoods, perform detector-level simulation, or derive new experimental confidence limits from raw event data.